PROJECT REPORT

FinAura Bot

AI-Powered Personal Financial Advisor

Submitted as part of the NASSCOM Virtual Internship Program
AI Specialist in BFSI

Submitted by
Antara Sonaje

Declaration

I, Antara Sonaje, submit this project report for the NASSCOM Virtual Internship Program in AI Specialist in BFSI. This report documents the FinAura Bot personal finance application as represented in the supplied website screenshots. The descriptions of features and workflows are based on the visible interface; implementation details not verifiable from the screenshots are presented as design intentions or recommendations rather than confirmed technical facts.

Acknowledgement

I would like to acknowledge the NASSCOM Virtual Internship Program in AI Specialist in BFSI for providing an opportunity to explore practical applications of artificial intelligence in financial services. I also acknowledge the guidance and learning resources that support the development of project-based skills in personal finance management, data visualization, and AI-assisted decision support.

Abstract

FinAura Bot is a personal financial advisory dashboard designed to help a user understand income, expenses, savings, budgets, and financial goals from one interface. The visible application includes a financial overview and scorecard, income and expense sections, an AI budget planner, savings goals, monthly reporting, and an AI chat assistant. The dashboard presents summary indicators such as monthly income, monthly expenses, net savings, financial health score, and emergency-fund runway. It also uses charts, category-level budget utilization, overspending alerts, and savings-goal progress bars to make financial information easier to interpret. This report describes the problem, objectives, main modules, sample dashboard data, expected workflow, benefits, limitations, and future improvements.

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Repository and Technical Architecture

Public repository: https://github.com/antarasonaje/personal-finance-advisor-bot. The repository README describes four seeded demonstration personas: a salaried professional, a college student, a freelance designer, and a household manager. The application is launched locally with `python app.py` and is documented to run at http://127.0.0.1:5000. A Gemini API key is optional; without it, the README says the app uses its built-in heuristic financial reasoning engine. Keep API keys and secret configuration out of public source control. The report does not claim that live bank-account integration is implemented.

The repository README lists a Python 3.10+ requirement, while also noting testing with Python 3.14. Because environment compatibility can vary, use the project's requirements.txt and verify the app in the target Python environment before demonstration.

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1. Introduction

Managing personal finances requires regular tracking of income, spending, savings, and upcoming goals. When this information is scattered across notes, spreadsheets, and separate applications, it can be hard to identify overspending or decide how much to save. FinAura Bot brings these activities into a single dashboard and presents them through summary cards, visual analytics, budget indicators, goal tracking, and an AI-oriented chat interface.

The project is relevant to the Banking, Financial Services and Insurance (BFSI) domain because it applies digital tools and AI-assisted guidance to a common financial-wellness use case. It is intended to support financial awareness and planning, not to replace a qualified financial adviser.

2. Problem Statement

Users may not have a clear, current view of their cash flow or category-wise spending. Without timely feedback, they can exceed budget limits, under-save for goals, or misjudge how long their emergency reserves might last. The project addresses this problem by consolidating financial indicators and highlighting areas that may need attention.

3. Objectives

- Present monthly income, expenses, net savings, and other financial indicators in one dashboard.
- Show spending patterns through daily trends and category-wise visualizations.
- Compare actual spending with allocated category budgets and flag overspending.
- Help users define savings goals, track progress, and record deposits toward those goals.
- Provide an AI chat interface for contextual financial questions and suggested actions.
- Generate a monthly view of financial activity that can support reflection and planning.

4. Scope of the Project

The visible scope covers personal budgeting and financial wellness. The interface shows a sample salaried profile, income and expense summaries, a budget planner, goal cards, a monthly report option, and a chat assistant. The screenshots do not establish whether the application connects to real bank accounts, uses live financial data, or persists information in a production database. Therefore, those capabilities are not claimed in this report.

5. System Overview and Modules

Module	Purpose
Overview & Scorecard	Displays headline indicators, daily spending trajectory, category breakdown, essential-versus-discretionary spending, and income-versus-expense comparison.
Incomes	Navigation section intended for reviewing or recording income sources. The supplied screenshots show the entry point but not the detailed form.
Expenses	Navigation section intended for recording and reviewing expenses. The overview provides a Log Expense action and category-level spending summaries.
AI Budget Planner	Shows category spend, allocated limits, utilization percentages, status labels, and remaining or over-budget amounts. It also presents controls to generate a budget plan or set custom category limits.
Savings Goals	Displays goal name, target date, target amount, saved amount, progress percentage, remaining amount, and a deposit action.

	Example goals include an emergency fund, a Japan vacation, and an investment portfolio.
Monthly Report	Navigation option for reviewing a period's financial performance. Detailed report contents are not fully visible in the supplied screenshots.
FinAura AI Chat	A chat interface presents a financial health assessment and suggested actions, with quick prompts for overspending analysis, 50/30/20 budgeting, emergency buffers, affordability checks, and reducing discretionary spending.
Active Persona and Reset Data	The header includes a persona selector and a Reset Data control. Reset should be used carefully because it may clear the current demonstration data.

6. Dashboard Analysis

6.1 Financial overview

The overview screenshot displays the following sample figures: monthly income of $5,800, monthly expenses of $4,255, net savings of $1,545, a financial health score of 67/100 marked Fair, and an emergency runway of 4.3 months. The displayed savings rate is 26.6%. The arithmetic is consistent for income minus expenses ($5,800 − $4,255 = $1,545). These are sample values shown in the interface, not verified real-world account data.

6.2 Spending and category analytics

A daily spending line chart illustrates changes across dates in September 2026. A category donut chart shows the distribution of spending across entertainment, food, healthcare, rent, shopping, transport, and utilities. A second pie chart compares essential needs and discretionary wants, while a bar chart compares monthly inflow, monthly outflow, and net savings. These visuals are intended to help users spot spending spikes and compare needs with wants.

6.3 Budget alerts

The interface displays an overspending alert indicating that entertainment is $116 above its monthly limit. The budget table also shows food as $139 over its limit and transport as $2 over its limit, while rent is near its limit at 97% utilization. Healthcare, shopping, utilities, and other categories are shown within their displayed allocations. The alert and table make the most urgent categories visible without requiring the user to inspect every transaction.

6.4 Savings goals

The savings-goal screen shows three example goals: an Emergency Fund with $9,800 saved toward $15,000 (65.3%); Vacation to Tokyo & Kyoto with $2,900 toward $4,500 (64.4%); and Stock Investment Portfolio with $5,400 toward $10,000 (54%). Each card shows a target date, remaining amount, progress bar, and deposit button. This design gives users a clear view of progress and the amount still required.

6.5 AI financial chat

The chat screen presents a financial health assessment with the sample score, cash flow, savings rate, essential/discretionary split, and emergency-buffer runway. It suggests reviewing overspent categories, checking savings goals, and asking questions such as how to optimize spending or generate a 50/30/20 budget. The screenshot indicates a context-aware assistant interface, but it does not independently prove the underlying AI model, data storage design, or accuracy of the advice.

7. User Workflow

14. Select the active persona shown in the header.
15. Review the overview cards to understand income, expenses, savings, health score, and emergency runway.
16. Inspect the charts to identify spending patterns and categories with high expenditure.
17. Open the budget planner and compare current spending with allocated limits.
18. Review savings goals and record a deposit when money is allocated to a goal.
19. Use the AI chat prompts to ask for explanations or budgeting suggestions.

20. Review the monthly report to reflect on the period and plan adjustments.

8. Technology and AI Considerations

The GitHub repository identifies the application as a Flask-based web application using SQLAlchemy for database interaction, with HTML templates, CSS styling, and JavaScript for client-side interaction and Chart.js visualizations. The repository includes app.py, config.py, database.py, models.py, a services folder, templates, static assets, requirements.txt, test_api.py, and verify_scenarios.py. The README describes a financial analytics service, an AI advisor service, and a seed service for sample personas. The AI layer is documented as supporting Google Gemini 2.5 Flash when a Gemini API key is configured, with a built-in heuristic reasoning fallback when no key is available. These details are based on the public repository README and file listing, not a full independent code audit.

For an AI-enabled BFSI application, good design should include clear explanations of recommendations, validation of numeric inputs, protection of personal financial data, safe handling of API keys, and a clear distinction between educational guidance and regulated financial advice. AI responses should be checked against the user's actual data and should not promise guaranteed investment outcomes.

9. Testing and Validation Plan

Test area	Validation
Dashboard calculations	Verify net savings equals income minus expenses and the savings rate uses a clearly defined formula.
Budget thresholds	Check that category status changes correctly when spending is below, near, or above the limit.
Goal progress	Confirm percentage and remaining amount calculations against saved and target values.
Input validation	Reject empty, malformed, negative, or otherwise invalid amounts where inappropriate.
AI chat	Test ordinary questions, missing context, contradictory inputs, and requests for unsafe or overconfident financial claims.
Reset behaviour	Confirm what data is cleared, warn the user before destructive reset, and verify the interface returns to a known state.
Responsive layout	Check navigation, charts, tables, and chat on desktop and mobile widths.
Privacy and security	Ensure secrets are not exposed in frontend code and personal financial data is not logged unnecessarily.

10. Benefits and Limitations

Benefits

- A consolidated view reduces the effort needed to understand monthly cash flow.
- Visual charts make trends and category distribution easier to interpret.
- Budget warnings help users identify overspending early.
- Goal progress indicators make savings targets concrete and trackable.
- The AI chat can provide a convenient conversational route to financial summaries and budgeting prompts.

Limitations

- The screenshots show sample data, so they do not prove connection to actual bank or payment systems.
- The accuracy and reliability of the AI advice cannot be evaluated from screenshots alone.
- The health score and emergency-runway calculations need documented formulas to be independently validated.
- A financial dashboard is only as accurate as its transaction data, budget settings, and calculation logic.
- The application should not be presented as a substitute for professional financial, tax, or investment advice.

11. Future Enhancements

- Add secure account authentication, role-based access, and consent-based data handling.
- Allow users to import transactions through a secure, clearly disclosed process and categorize them automatically.
- Explain the formula behind each score, alert, and recommendation.
- Add recurring expense detection, bill reminders, and cash-flow forecasting.
- Support exportable monthly reports and accessible chart descriptions.
- Provide multilingual support and mobile-first responsive layouts.
- Add audit logs, error handling, and tests for financial calculations.
- Include disclaimers and escalation guidance for decisions that require a qualified financial professional.

12. Conclusion

FinAura Bot demonstrates a useful personal-finance dashboard concept for the BFSI domain. Its main strengths are the combination of financial summaries, visual spending analytics, category budgets, savings-goal tracking, and a conversational AI interface. The screenshots show how these features can help a user notice budget overruns and monitor progress toward financial goals. Before deployment with real users or real financial data, the application should be validated for calculation accuracy, privacy, security, explainability, and the quality of its AI-generated guidance.

Appendix A: Website Screenshots

The following screenshots were supplied for the report and are included as visual evidence of the application interface.

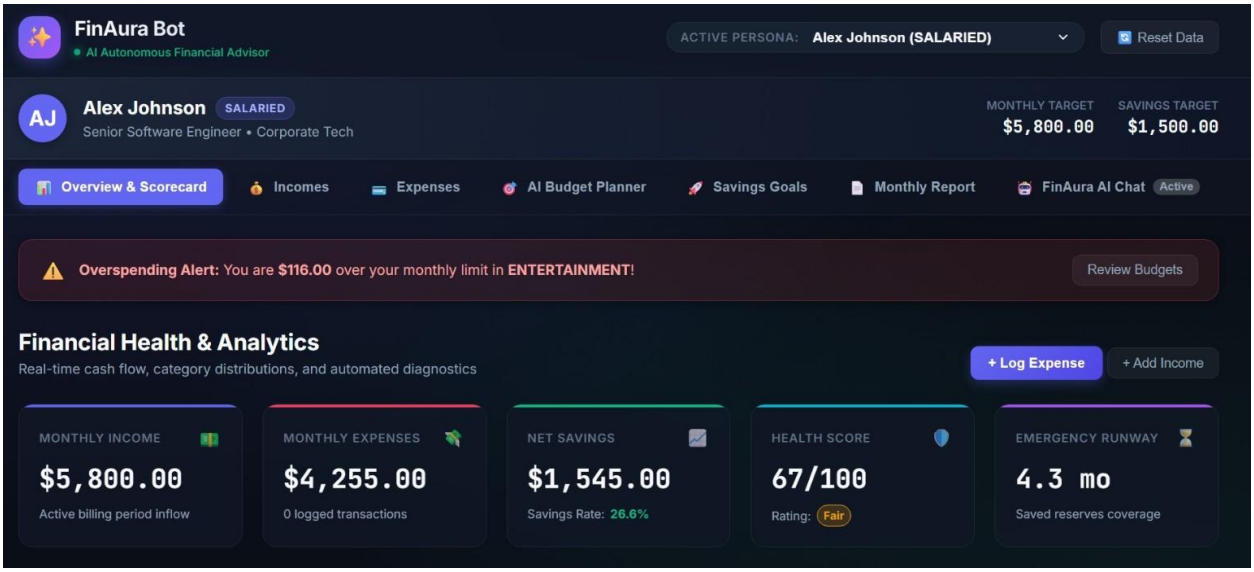


Figure A1. Financial overview and analytics

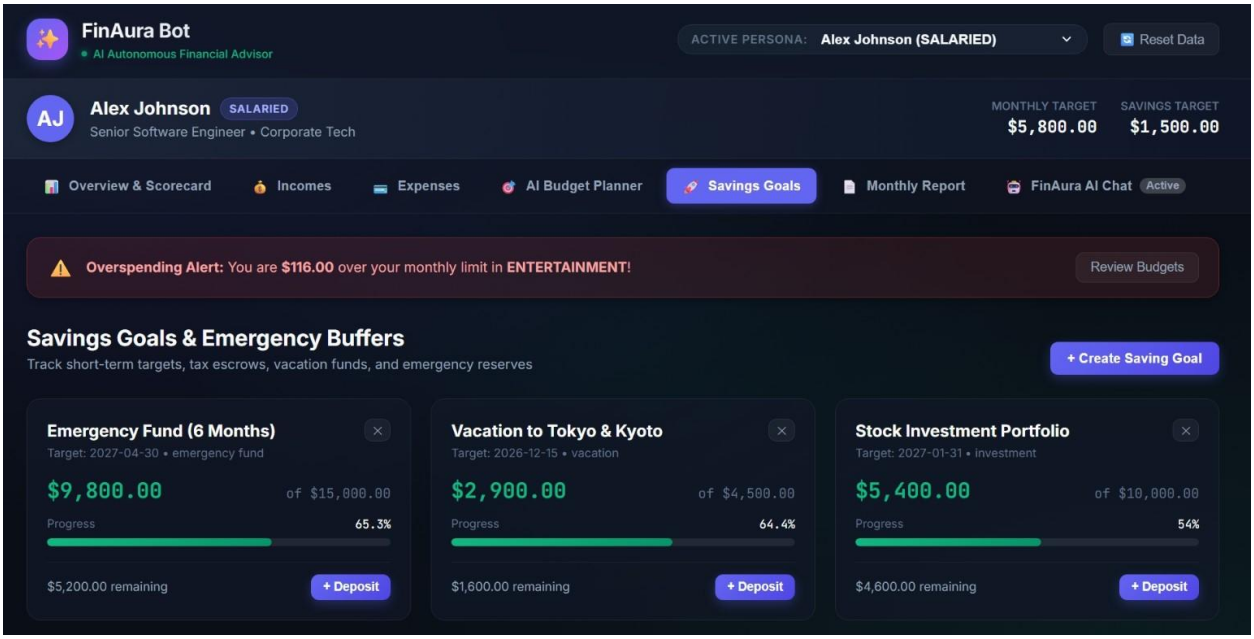


Figure A2. Savings goals and emergency buffers

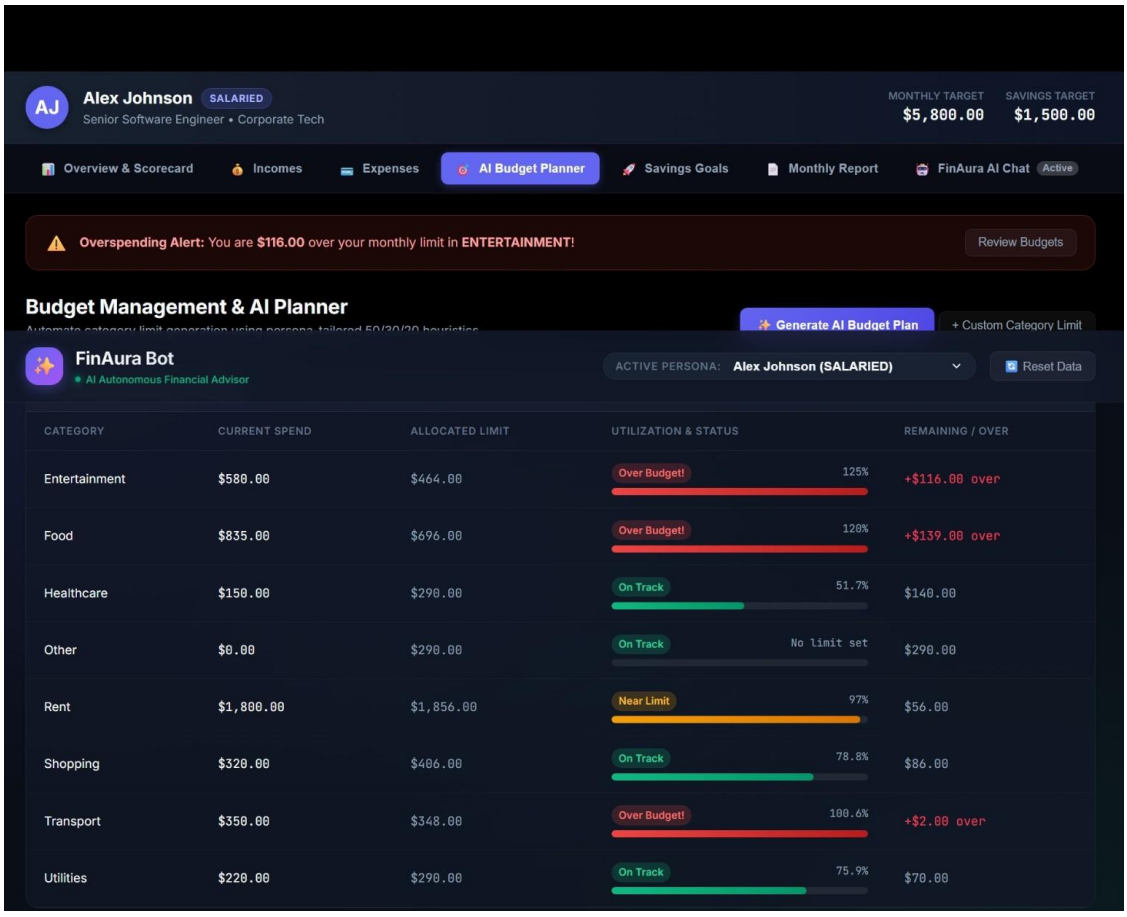


Figure A3. AI budget planner and category utilization

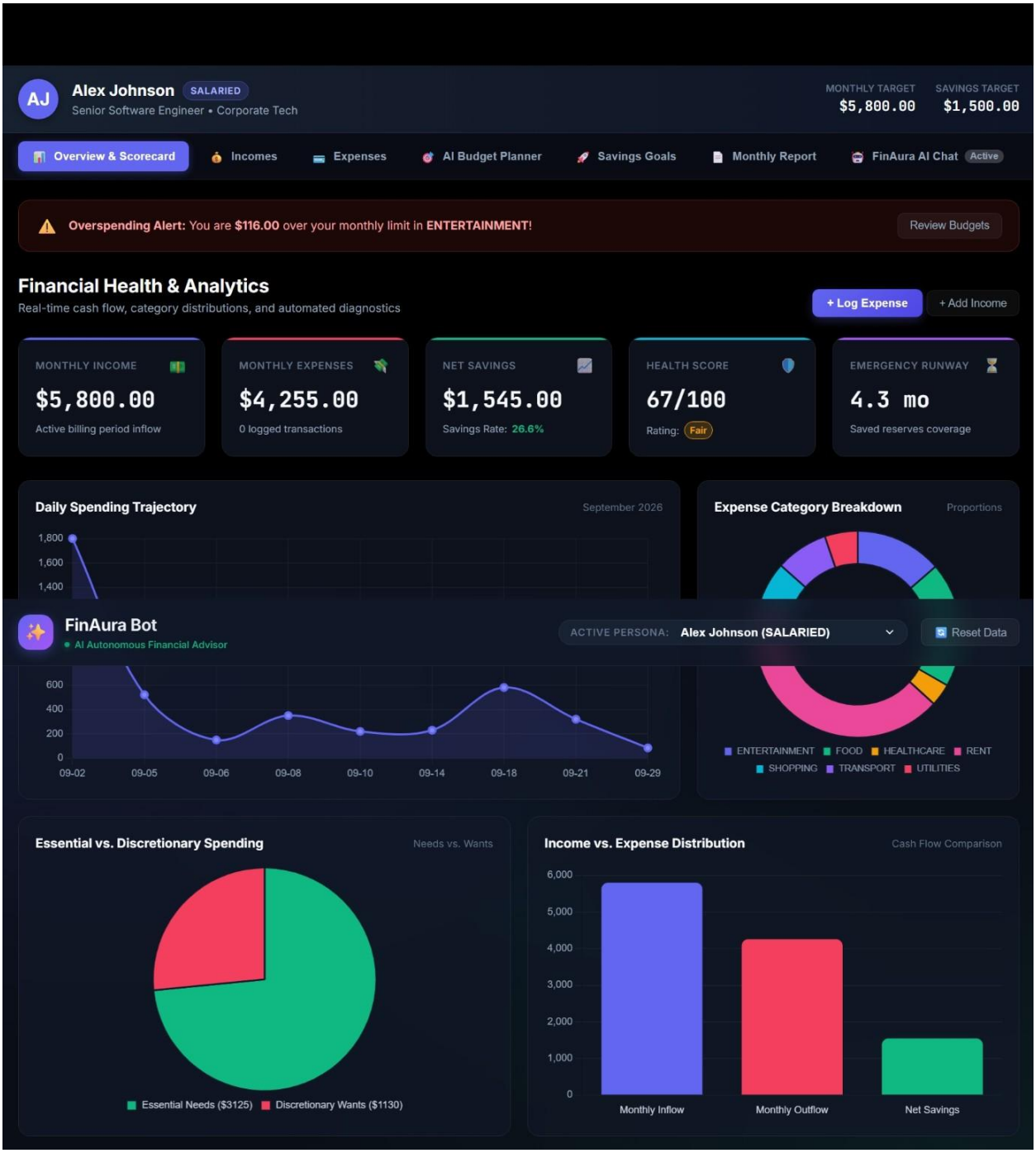


Figure A4. Overview scorecard and key financial indicators

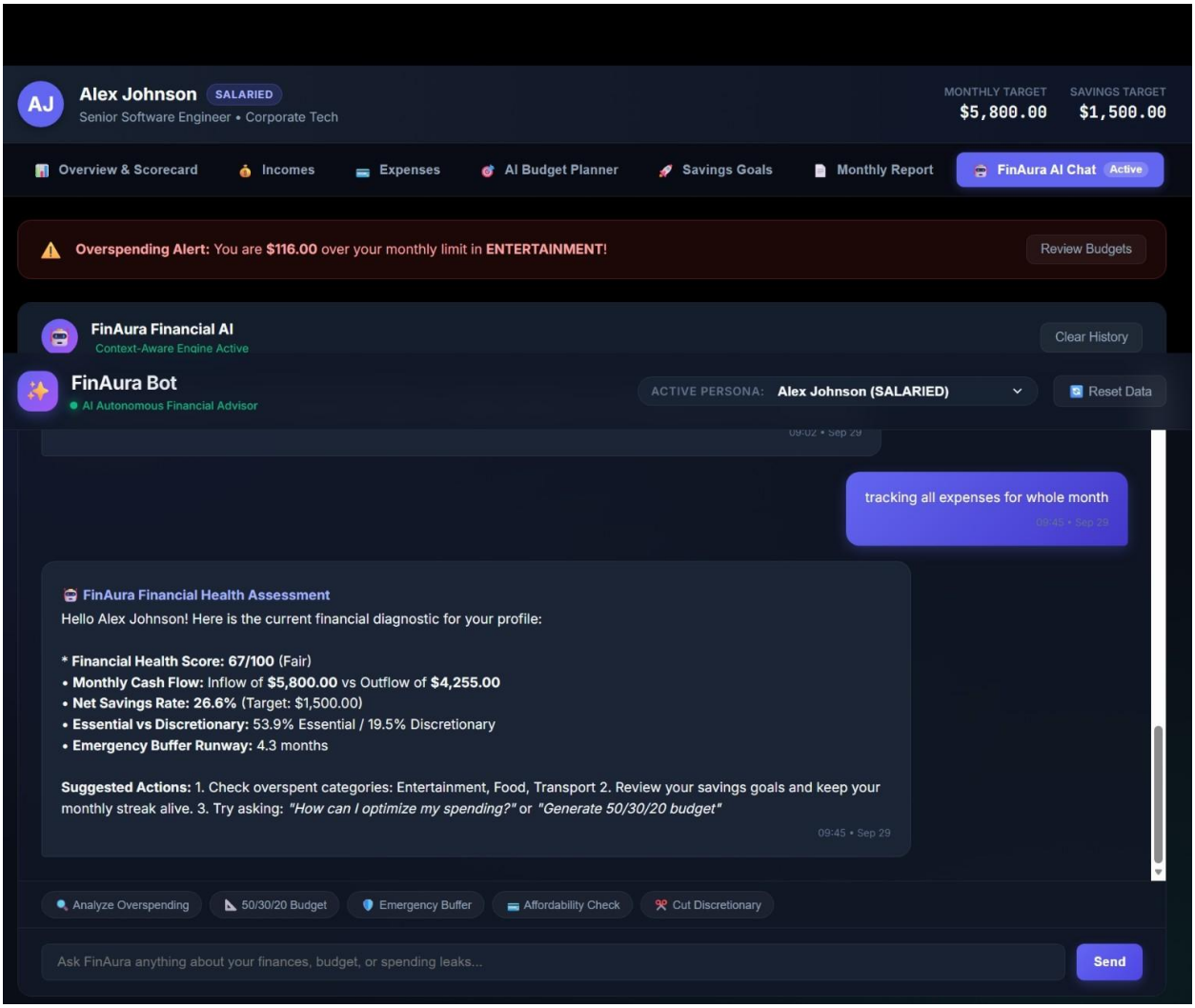


Figure A5. FinAura AI chat and financial health assessment